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42088 – Industrial Project

Project Vision

Project name:	Automatic, universal push-button actuator, equipped with force gauge
Customer:	Altice Labs
Team members:	Coordinator: Cláudia Almeida, claudiasalmeida@ua.pt, 914102617. Other team members: Rafael Alves, rafael01@ua.pt, 937353627. Gonçalo Peralta, goncaloperalta@ua.pt, 916828627. Pedro Matias, pedrodsmatias@ua.pt, 917126548.
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Revision History

Date	Issue	Description	Author
22/10/2024	Manual button testing approach	Exploring the problem, outlining a solution, assessing possible risks	Eng. Hélder Alves

1 Project description

The goal of this project is to automate the process of pressing buttons, checking their operation, and determining the force required to activate them. Traditionally, button testing has been done manually, which requires the presence of a person at the test site, making remote actuation of the buttons by the automatic tests unfeasible. Additionally, manual actuation can make it difficult to perceive when the button clicks and when it does not, especially when there are many buttons to press in sequence. Pressure force appears as another variable that is difficult to measure accurately by humans, as pass/fail criteria will be subjective from person to person. Automating this operation improves precision, reproducibility, and dependability, particularly in circumstances where many buttons of various sorts and sizes must be examined. Our proposed solution fits the needs of Altice Labs and opens the door to broader industrial applications by being universally adaptable to varied devices under test (DUT). This project addresses the growing demand for automation in testing processes across sectors. This automated technology has major advantages, including increased testing speed, greater data accuracy, and the ability to remotely control and monitor the testing process. As a result, it will improve the workflow for firms like Altice Labs, which test network gateways, routers, and other devices with physical interfaces lowering labour costs, enhancing product reliability and test more software releases in less time.
The system guarantees that force is applied consistently, yielding precise and reproducible findings, which is critical for determining if devices satisfy the required requirements. Furthermore, it is intended to accommodate a wide number of devices by adapting to different button sizes and mechanisms, making it applicable to a diverse range of products. The capacity to collect detailed force measurements and process information will also allow businesses to make data-driven decisions that will advise design changes, increase product reliability, and direct manufacturing adjustments.

2 Deliverables/Outcome

The tangible deliverables will feature a fully functional prototype of the automated button actuator system, controlled via an API and capable of pressing various types of buttons with adjustable force. This system will incorporate a linear actuator for precise movement, a resistive force sensor for measuring applied force, and custom-designed accessories (potentially 3D-printed) to ensure compatibility with different devices. Additionally, a complete software solution will be developed to remotely manage the actuator and collect data, transmitting test results to a server for analysis. Figure 1 shows a practical sketch of the automaton to be built.

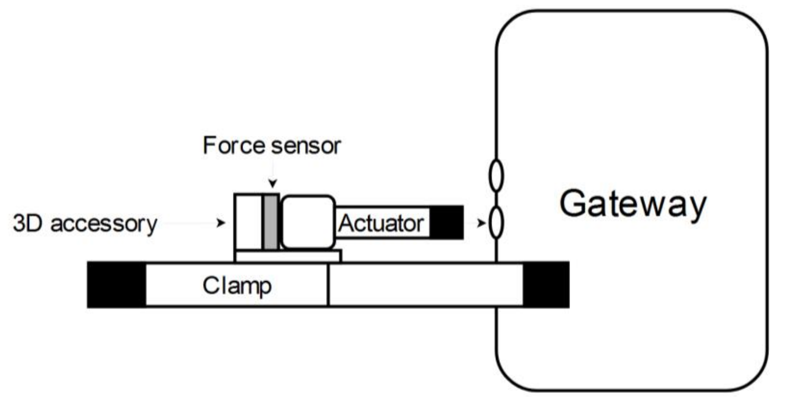

Figure 1. Global design of the automaton.

On the intangible side, the project will generate valuable technical expertise gained in designing and building automated testing systems, integrating actuators and sensors with microcomputers like the

Raspberry Pi as well as knowledge of force measurement techniques and how to apply them in real-world testing environments.

3 Functionality and Technology

The system will automate the task of pressing buttons on various devices, such as network gateways and routers, while measuring the force required to activate each button. The interaction between users, primarily the test engineers at Altice Labs, and the final product will be largely remote rather than physical. The system is designed to be easily applied to the DUT without the need for manual adjustments, such as fine-tuning the height or positioning relative to the buttons being tested. Instead, users will interact with the system through an intuitive web interface which will display all test history stored in the database. This interface will deliver the main advantage of the project: providing real-time statistics on the force applied during tests, as well as feedback on the functionality of the buttons by allowing to initiate button tests remotely from any connected device, adjust test parameters, including the amount of force to apply, and select the type of DUT.

To achieve the system's functionality, several key technologies will be used. A **Raspberry Pi** will act as the central controller, managing communication between the linear actuator, force sensor, and server. The **linear actuator** will perform the button pressing, offering precise movement and force control, making it adaptable to various button types. A **resistive force sensor** will measure the applied force during each press, with data converted by an **Analog to Digital Converter (ADC)** for processing by the Raspberry Pi, ensuring accurate force readings. The system will also feature an **API and web interface** to enable remote control and automation of tests, allowing users to adjust settings and retrieve data easily. Additionally, **custom 3D-printed fixtures** will be designed/adapted to hold the actuator and DUTs securely, ensuring a universal fit for a variety of devices.

The system will operate within the testing labs of Altice Labs, where test engineers can physically set up the DUTs. However, the remote capabilities of the system, enabled through the web interface and API, will allow users to control the testing process from any location connected to the lab network.

4 Customer / End users / Market

The primary target user of our project is the test engineering team at Altice Labs, who will directly use the automated button actuator system to test devices such as routers and gateways. These engineers are responsible for ensuring the functionality and reliability of these devices before they are released to the market. While they will operate the system, the stakeholders extend beyond just the test engineers. The product development team will benefit from the accurate and consistent testing data that can be used to improve product design, while the quality assurance (QA) team will rely on the system’s results to verify that devices meet required specifications before production.
In a further sense, Altice Labs' management team is also a stakeholder, as automation will increase efficiency, reduce testing time and potentially lower labour costs, aligning with the company's objectives of improving productivity and streamlining operations.

To measure customer satisfaction, several key factors can be considered once the product is in use:

- Efficiency gains: The improvement in software testing time during daily software releases, through the inclusion of button tests in current automatic software tests, thus increasing the automatic tests coverage, eliminating the need for manual tests on buttons.
- Accuracy: The precision and consistency of force measurements and button functionality results.
- Usability: Feedback from test engineers on the intuitiveness and user-friendliness of the system.

- Reliability: The system's ability to handle various types of devices consistently, without failures or frequent adjustments.

By assessing these factors, we can ensure that the system meets the needs of its direct users, while providing value to all the stakeholders involved in the project.

5 Partners

This project has a key institutional partner: **Altice Labs**, a leading technology company focused on innovation and development in telecommunications and IT solutions. Altice Labs plays a crucial role as both the client and collaborative partner in this project.
Its main role includes:

- Providing the problem statement: Altice Labs has defined the scope and objectives of the project, aiming to automate the testing of devices such as routers and gateways by developing an automated button actuator.

- Technical Support and Resources: Through offering access to their workspaces, equipment, and test devices such as the Raspberry Pi and DUTs (devices under test). This support allows us to carry out preliminary tests.

- Component Sourcing: Altice Labs will assist with the acquisition of necessary components, ensuring that there are no delays in delivery, thus helping us stay on schedule.

- Ongoing Feedback: Throughout the project, engineers from Altice Labs, such as Eng. Hélder and Eng. Nuno, are actively involved in providing technical feedback and guidance. They will help validate key design decisions and ensure that the final product meets their operational requirements.

In addition to the company that proposed the challenge, we are fortunate to have the guidance and support of **Professor Pedro Fonseca**, from the Department of Electronics and Telecommunications at the University of Aveiro. He will be closely involved throughout the entire project, providing mentorship and oversight from the initial planning stages through to the final proof of concept.